

Variables: Java

boolean Data Type

In Java, the `boolean` primitive data type is used to store a value, which can be either `true` or `false`.

```
boolean result = true;
boolean isMarried = false;
```

Strings

A String in Java is a Object that holds multiple characters. It is not a primitive datatype.

A String can be created by placing characters between a pair of double quotes (`"`).

To compare Strings, the `equals()` method must be used instead of the primitive equality comparator `==`.

```
// Creating a String variable
String name = "Bob";

// The following will print "false"
// because strings are case-sensitive
System.out.println(name.equals("bob"));
```

int Data Type

In Java, the `int` datatype is used to store integer values. This means that it can store all positive and negative whole numbers and zero.

```
int num1 = 10;    // positive value
int num2 = -5;    // negative value
int num3 = 0;     // zero value
int num4 = 12.5;  // not allowed
```

char Data Type

In Java, `char` is used to store a single character. The character must be enclosed in single quotes.

```
char answer = 'y';
```

Primitive Data Types

Java's most basic data types are known as *primitive data types* and are in the system by default.

The available types are as follows:

- `int`
- `char`
- `boolean`
- `byte`
- `long`
- `short`
- `double`
- `float`

`null` is another, but it can only ever store the value `null`.

```
int age = 28;

char grade = 'A';

boolean late = true;

byte b = 20;

long num1 = 1234567;

short no = 10;

float k = (float)12.5;

double pi = 3.14;
```

Static Typing

In Java, the type of a variable is checked at compile time. This is known as *static typing*. It has the advantage of catching the errors at compile time rather than at execution time.

Variables must be declared with the appropriate data type or the program will not compile.

```
int i = 10;           // type is int
char ch = 'a';        // type is char

j = 20;               // won't compile, no
                      // type is given
char name = "Lil";    // won't compile,
                      // wrong data type
```

final Keyword

The value of a variable cannot be changed if the variable was declared using the `final` keyword.

Note that the variable must be given a value when it is declared as `final`. `final` variables cannot be changed; any attempts at doing so will result in an error message.

```
// Value cannot be changed:
final double PI = 3.14;
```

double Data Type

The `double` primitive type is used to hold decimal values.

```
double PI = 3.14;  
double price = 5.75;
```



Print